

A Centered Counterexample to the Orlicz-Supremum Question

Full negative answer to arXiv:0808.3248, equation (14)

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Status. `candidate_counterexample_likely_valid`. The answer to the source question is negative, even for a centered field on a countable index set.

Source question

Ostrovsky asks on p. 6 of arXiv:0808.3248 whether every separable random field $\theta = (\theta(t))_{t \in T}$ satisfying

$$\sup_{t \in T} |\theta(t)| < \infty \text{ a.s.}, \quad \sup_{t \in T} \|\theta(t)\|_{\text{Or}(\Phi)} < \infty$$

must also satisfy

$$\left\| \sup_{t \in T} |\theta(t)| \right\|_{\text{Or}(\Psi)} < \infty \tag{14}$$

whenever $\Psi \ll \Phi$, meaning $\Psi(vu)/\Phi(u) \rightarrow 0$ as $u \rightarrow \infty$ for every $v > 0$.

$$|\zeta(t) - \zeta(s)| \leq \zeta \times r(t, s),$$

where

$$\mathbf{E}\Psi(\zeta) < \infty.$$

DH. In the connection of the last assertion we dare formulate the following, interest by our opinion, hypothesis. Let $\theta = \theta(t)$, $t \in T$ be arbitrary separable random field, centered: $\mathbf{E}\theta(t) = 0$ or not, bounded with probability one:

$$\sup_{t \in T} |\theta(t)| < \infty \text{ a.e..}$$

Assume in addition that for some Orlicz function $\Phi(\cdot)$

$$\sup_{t \in T} \|\theta(t)\|_{\text{Or}(\Phi)} < \infty.$$

Let also Ψ be another Orlicz function such that $\Psi \ll \Phi$.

Open question: there holds (or not)

$$\left\| \sup_{t \in T} |\theta(t)| \right\|_{\text{Or}(\Psi)} < \infty? \tag{14}$$

The conclusion (14) is true for the *centered* Gaussian fields [8]; if the field $\theta(\cdot)$ satisfies the so - called entropy or generic chaining conditions [5], [9], [10]; in the case if θ belongs to the domain of attraction of Law of Iterated Logarithm [11] etc.

Finally, let us consider the following example. Let τ be Normal (Gaussian) standard distributed r.v.: $\text{Law}(\tau) = N(0, 1)$ ant let $T = R = (-\infty, \infty)$. We define

$$\theta(t) = \tau \cdot \exp\left(-\frac{t^2}{2}\right).$$

For an Orlicz function F , we use the source's Luxemburg norm

$$\|X\|_{\text{Or}(F)} = \inf\{c > 0 : \mathbf{E}F(|X|/c) \leq 1\}.$$

Proof intuition

Uniform control of each coordinate says nothing about how many disjoint rare coordinates occur at a given amplitude. We split a level of total probability 2^{-m} into roughly e^{2m^4} separate coordinates of amplitude m^2 . Each coordinate then has a negligible Gaussian-Orlicz moment. A sample point activates only one coordinate, so every sample path is bounded. The supremum, however, forgets the splitting: it has size comparable to m^2 with total probability 2^{-m} and therefore has no exponential moment at any scale. Subtracting each atom's probability centers every coordinate without changing this mechanism.

Theorem 1. *Let*

$$\Phi(u) = e^{u^2} - 1, \quad \Psi(u) = e^{|u|} - 1.$$

There is a centered separable real random field $\theta = (\theta(t))_{t \in T}$ such that

$$\sup_{t \in T} \|\theta(t)\|_{\text{Or}(\Phi)} \leq 2, \quad \sup_{t \in T} |\theta(t)| < \infty \quad a.s.,$$

but

$$\left\| \sup_{t \in T} |\theta(t)| \right\|_{\text{Or}(\Psi)} = \infty.$$

Moreover T is countable. Hence equation (14) is false, including under the optional centering hypothesis in the source.

Proof. For $m \geq 1$, put

$$q_m = 2^{-m}, \quad a_m = m^2, \quad N_m = \left\lceil e^{2m^4} \right\rceil, \quad p_m = \frac{q_m}{N_m}.$$

Take a countable atomic probability space whose atoms are $E_{m,j}$, $m \geq 1$, $1 \leq j \leq N_m$, with $\mathbb{P}(E_{m,j}) = p_m$. This is a probability space because

$$\sum_{m \geq 1} \sum_{j=1}^{N_m} p_m = \sum_{m \geq 1} q_m = 1.$$

Let $T = \{(m, j) : m \geq 1, 1 \leq j \leq N_m\}$ and define

$$\theta(m, j) = a_m(\mathbf{1}_{E_{m,j}} - p_m). \tag{1}$$

The index set is countable, so the field is separable, and $\mathbb{E}\theta(m, j) = 0$ for every coordinate.

We first verify the uniform Φ bound. Since $N_m \geq e^{2m^4}$,

$$p_m \leq 2^{-m} e^{-2m^4}, \quad a_m p_m \leq \frac{1}{2}. \tag{2}$$

At Luxemburg scale 2, using $1 - p_m \leq 1$ and (2),

$$\begin{aligned} \mathbb{E}\Phi\left(\frac{|\theta(m, j)|}{2}\right) &= p_m \Phi\left(\frac{a_m(1 - p_m)}{2}\right) + (1 - p_m) \Phi\left(\frac{a_m p_m}{2}\right) \\ &\leq p_m e^{a_m^2/4} + \Phi(1/4) \\ &\leq \frac{1}{2} e^{-7/4} + (e^{1/16} - 1) < 1. \end{aligned}$$

Thus $\|\theta(m, j)\|_{\text{Or}(\Phi)} \leq 2$ uniformly.

Next fix an outcome $\omega \in E_{M,J}$. Its distinguished coordinate has absolute value $a_M(1 - p_M) \leq a_M$, while every other coordinate has absolute value $a_m p_m < 1$ by (2). Hence

$$\sup_{t \in T} |\theta(t, \omega)| \leq \max\{a_M, 1\} < \infty.$$

The sample paths are therefore bounded almost surely. On the same atom, $p_M < 1/2$ and the distinguished coordinate also gives the lower bound

$$S(\omega) := \sup_{t \in T} |\theta(t, \omega)| \geq a_M(1 - p_M) \geq \frac{M^2}{2}. \quad (3)$$

For any $c > 0$, summing (3) over all atoms at level m yields

$$\begin{aligned} \mathbb{E}\Psi(S/c) &\geq \sum_{m \geq 1} \sum_{j=1}^{N_m} p_m \Psi\left(\frac{m^2}{2c}\right) \\ &= \sum_{m \geq 1} 2^{-m} \left(e^{m^2/(2c)} - 1\right) = \infty. \end{aligned}$$

Indeed, the summands tend to infinity because $m^2/(2c) - m \log 2 \rightarrow \infty$. No Luxemburg scale is admissible, so $\|S\|_{\text{Or}(\Psi)} = \infty$.

Finally, $\Psi \ll \Phi$: for every fixed $v > 0$,

$$\frac{\Psi(vu)}{\Phi(u)} = \frac{e^{vu} - 1}{e^{u^2} - 1} \rightarrow 0.$$

All hypotheses of the source question hold and its conclusion fails. □

Verification and limitations

The proof is exact and has no computational dependency. The total atom mass, centering, countable separability, pathwise boundedness, uniform Luxemburg bound, weakening relation, and divergence at every scale were audited separately in `verification_report.md`.

The construction is deliberately non-Gaussian and highly nonstationary. It does not contradict the positive Gaussian or generic-chaining cases cited by the source. The example printed immediately after the source question controls only $\sup_t \theta(t)$; its $\sup_t |\theta(t)|$ is infinite. Construction (1) meets the absolute-boundedness assumption exactly.

Novelty check and review recommendation

Bounded searches used the exact title and arXiv id, verbatim question fragments, and combinations of “uniform Orlicz norm,” “random field,” and “supremum counterexample.” They found the source and general sufficient- condition literature but no later exact answer. This supports, but does not certify, novelty. Human review should focus on the scale-2 moment estimate and the aggregation from coordinate mass p_m to level mass $q_m = 2^{-m}$.

References

- [1] E. Ostrovsky, *Support of Borelian Measures in Separable Banach Spaces*, arXiv:0808.3248, 2008, p. 6, equation (14).